

KAUSHIK MASLEKAR

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<https://github.com/kaushikkishormaslekar> portfolio: kaushikmaslekar.vercel.app

SUMMARY

AI Systems Engineer focused on building intelligent backend applications using Retrieval-Augmented Generation (RAG), semantic search, and scalable API-driven architectures. Experienced in vector search systems, backend integration, data preprocessing, and distributed application development using Python and Java technologies. Strong interest in modern AI infrastructure, cloud-native systems, and production-grade AI workflows.

TECHNICAL SKILLS

AI/ML - RAG, Semantic Retrieval, Embeddings, Pinecone, ChromaDB, Vector Search, Pandas, NumPy
Backend - Java, Spring Boot, FastAPI, REST APIs, Microservices, Kafka, Redis, Caching
Cloud & DevOps - AWS, Docker, Kubernetes, CI/CD, Prometheus, Grafana
Databases - MySQL, MongoDB, Redis
Hardware & IoT - ESP32, Sensor Integration, Embedded Systems Basics, IoT Monitoring Systems, Computer Networking

EXPERIENCE

- AI/ML Intern | YBI Foundation | Feb 2026 – Mar 2026
- Built a Retrieval-Augmented Generation (RAG) application using Python, LangChain, and vector embeddings to enable document-grounded question answering.
 - Implemented document chunking, embedding generation, and semantic retrieval mechanisms to improve response accuracy and knowledge retrieval performance.
 - Performed exploratory data analysis, statistical validation, and data cleaning to prepare datasets for machine learning experimentation.
 - Collaborated on practical AI solutions involving LLM applications, information retrieval, and end-to-end data processing pipelines.

KEY PROJECTS

Real-Time Payment Processing System
Designed and implemented a scalable payment backend using Spring Boot, Kafka, PostgreSQL, and Redis. Implemented idempotency keys, retry mechanisms, and asynchronous event-driven workflows to ensure reliable transaction processing and audit logging.
Technologies: Java, Spring Boot, Kafka, PostgreSQL, Redis, Docker, AWS

Knowledge-Based Search Assistant (RAG)
Built a Retrieval-Augmented Generation (RAG) application for semantic search and contextual question answering over large document collections. Implemented document chunking, embedding generation, and similarity-based retrieval using vector databases.
Technologies: Python, LangChain, FAISS, Vector Databases, RAG, LLM Applications

Distributed API Rate Limiter
Built a distributed rate-limiting service using Spring Boot and Redis, implementing the Token Bucket algorithm to enforce per-user and per-IP request limits. Designed the system to protect backend APIs from excessive traffic while maintaining low-latency performance.
Technologies: Java, Spring Boot, Redis, Docker

EDUCATION

B.E. in Artificial Intelligence and Data Science (Pursuing)
Marathawada Mitra Mandal's College of Engineering, Pune | Aug 2024 – Present